Natural Selection I: Patterns and Process

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Intro Bio: Ecology and Evolution

Opening Question

What is the difference between ‘evolution’ and ‘natural selection’?

A. Nothing, these terms are interchangeable
B. Natural selection is short-term change (allele frequencies), evolution is long-term change (speciation)
C. Evolution is any kind of genetic change, natural selection is a particular ‘type’ of change (adaptation)
D. Natural selection produces analogous traits, evolution produces homologous traits
E. I think they’re different, but I’m not sure how

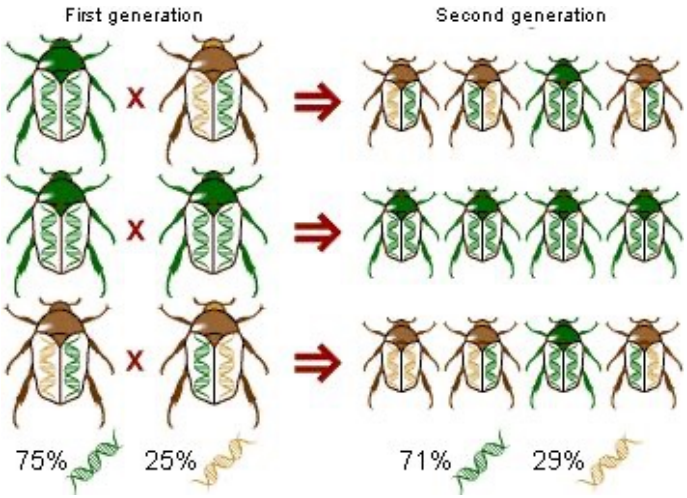
Learning Objectives

By the end of this lecture, students should be able to…

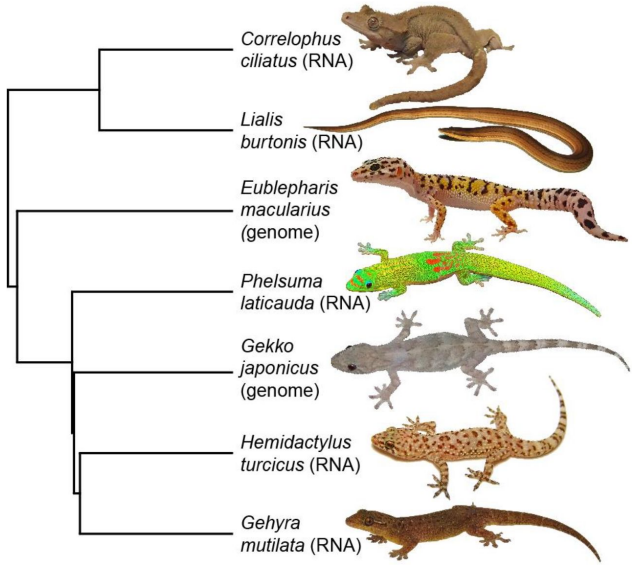
- Determine if a population meets the conditions necessary for natural selection to occur.
- Predict the outcome of natural selection within a given scenario (including different drivers, levels of heritability, fitness functions, etc)
- Identify the factors that determine the relative fitness of different phenotypes in a given scenario

Evolution: Genetic Change

- Microevolution



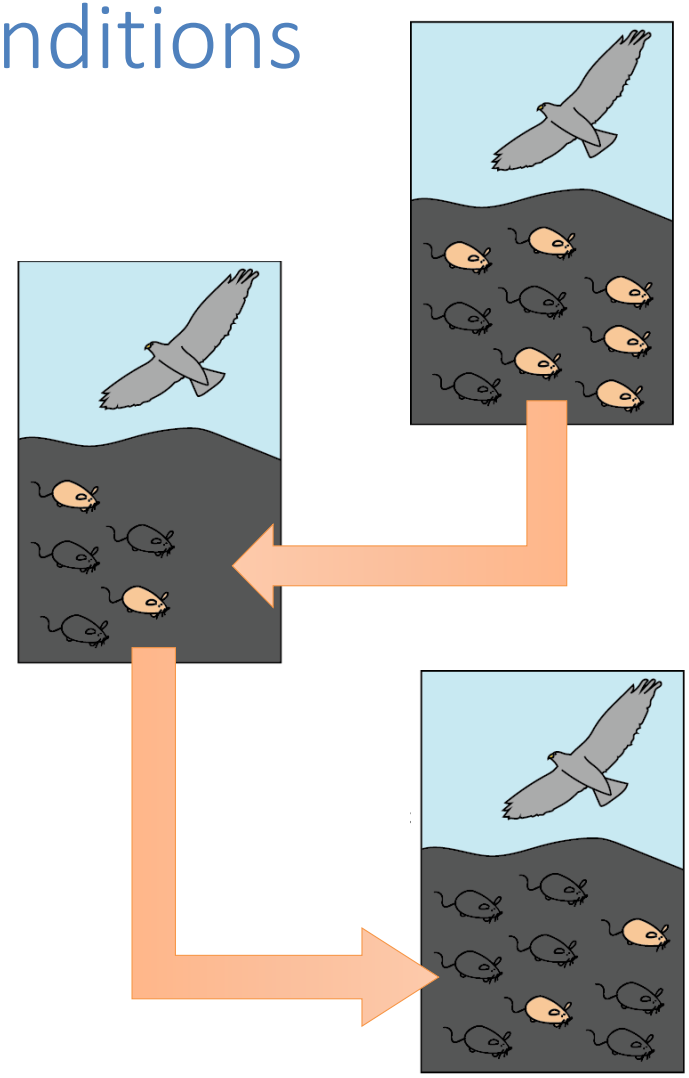
- Macroevolution



Necessary and Sufficient Conditions

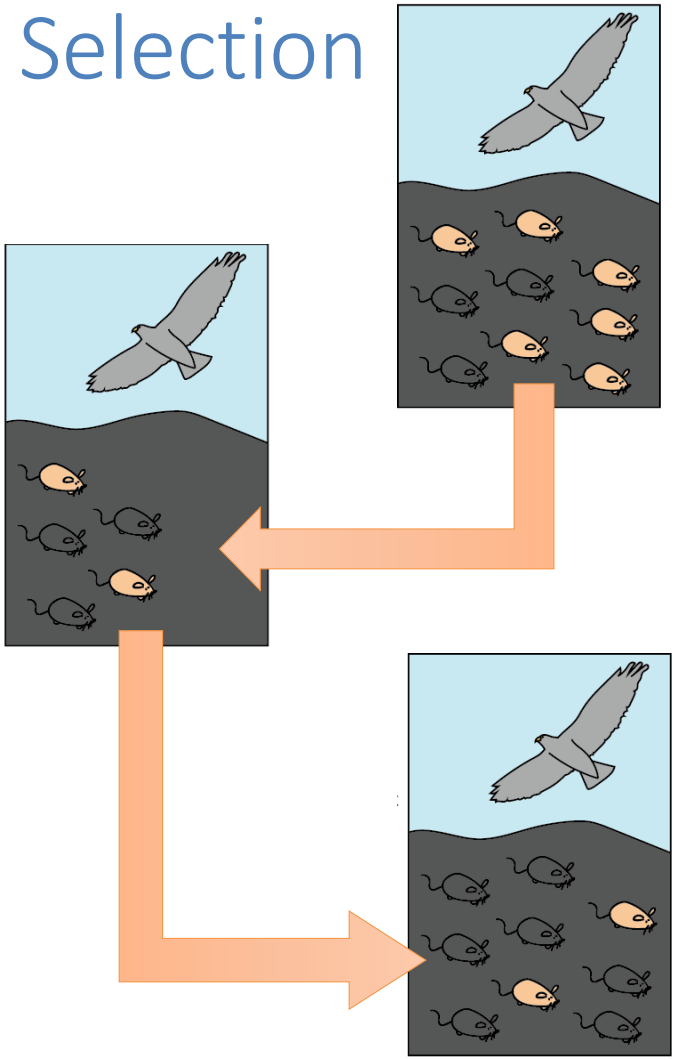
Natural Selection: Increased survival and reproduction of some individuals compared to others, based on differences in their traits (phenotypes).

- Variation
  - Individuals vary in the trait (phenotype)
- Heritability
  - The trait is at least partially genetic; individuals pass on the trait to their offspring
- Differential Success
  - Individuals with different traits differ in their survival or reproductive success (fitness)



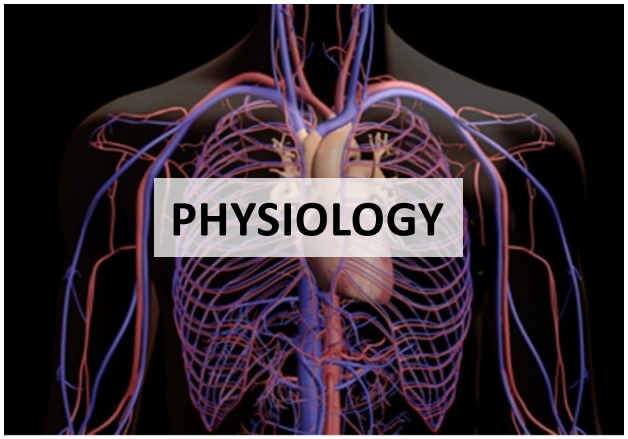
Patterns and Drivers of Natural Selection

- Pattern: the impact that selection has on frequencies of phenotypes
  - How will the frequency of tan and black mice change across generations?
- Driver: the factor that determines which phenotype(s) has higher fitness?
  - Why do black mice have higher fitness in this environment?



1. Phenotypic variation

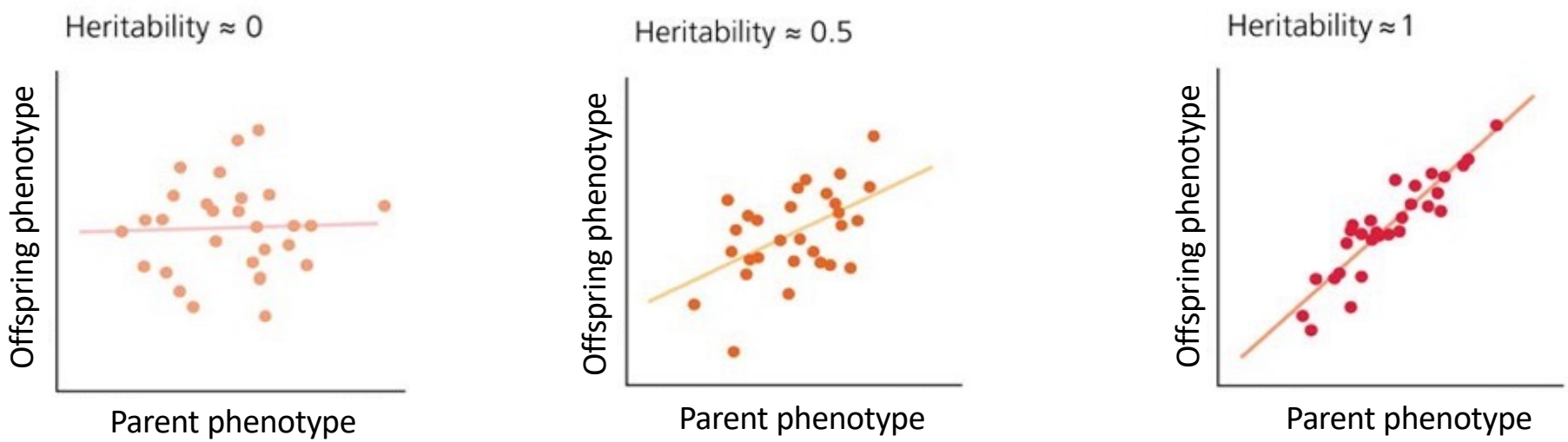
Phenotype: A set of observable characteristics of an organism including:



2. Heritability

The proportion of trait variation explained by inheritance; the degree to which parental phenotypes predict offspring phenotypes

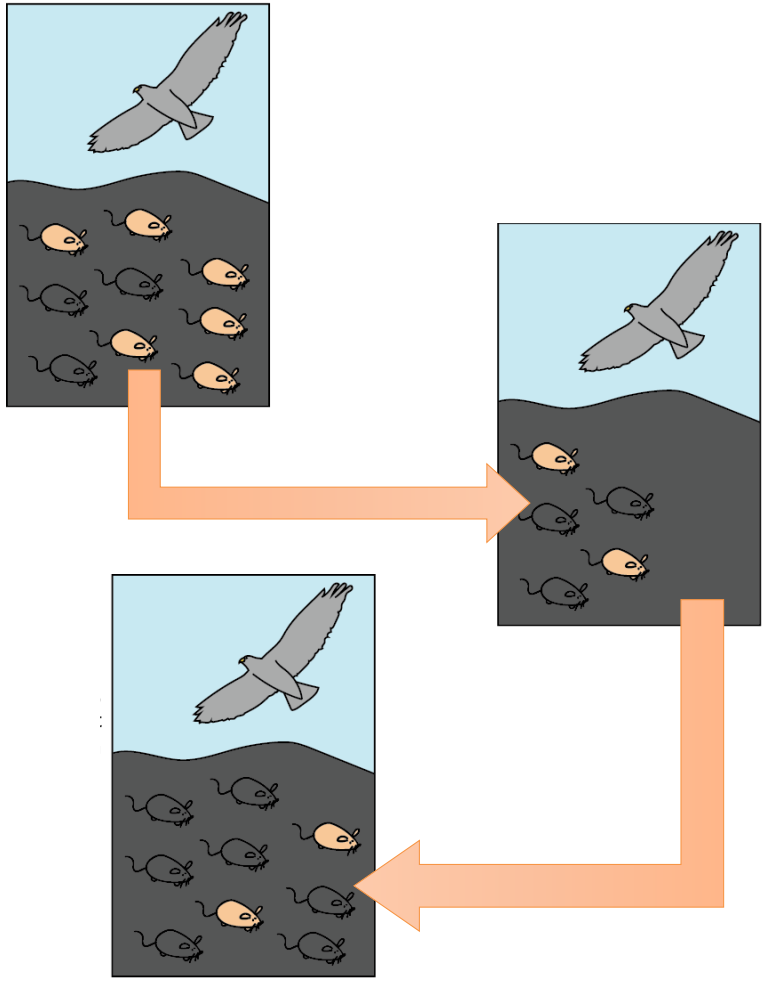
- 0 = not genetic (completely environmental)
- 1 = completely genetic (no environment effect)



3. Differential Success

An organism's ability to survive and reproduce in a particular environment

Generation 1	Generation 2	Fitness (# offspring)
		1
		3
		0

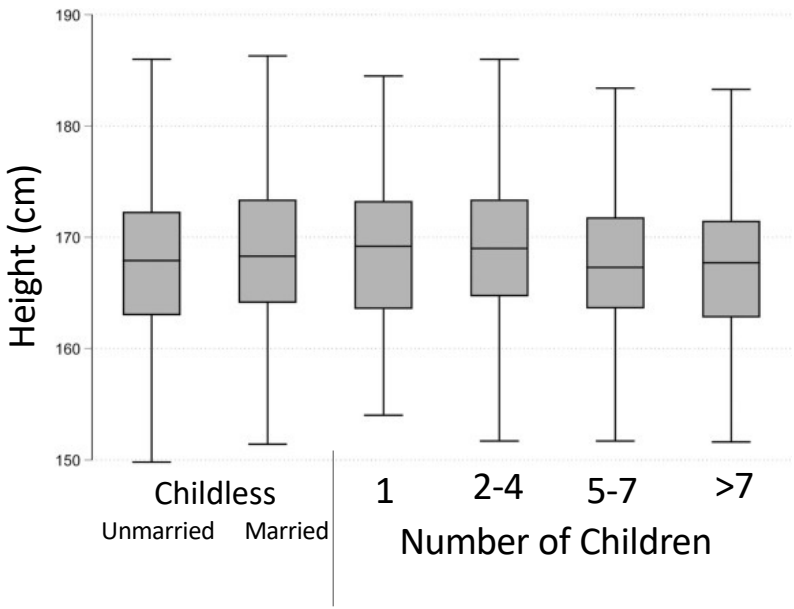


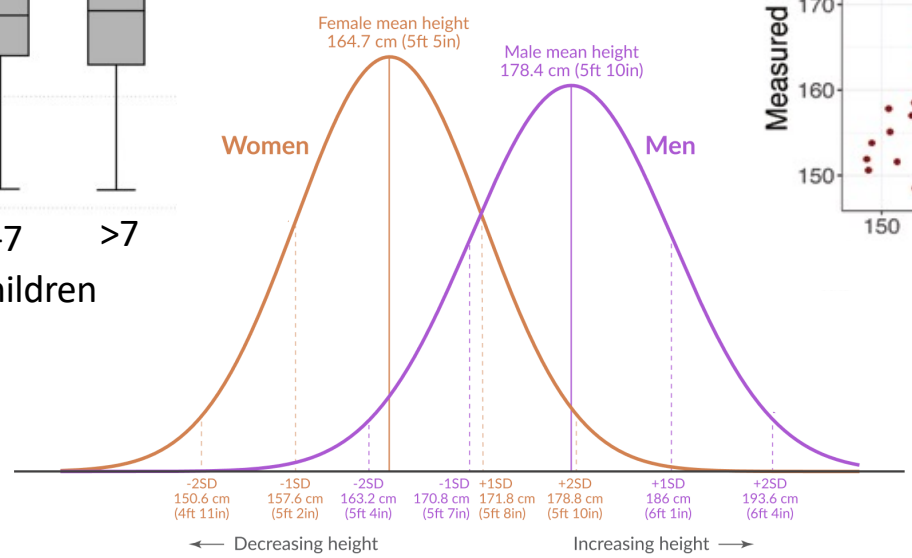
Do we have evidence for each of the necessary and sufficient conditions?

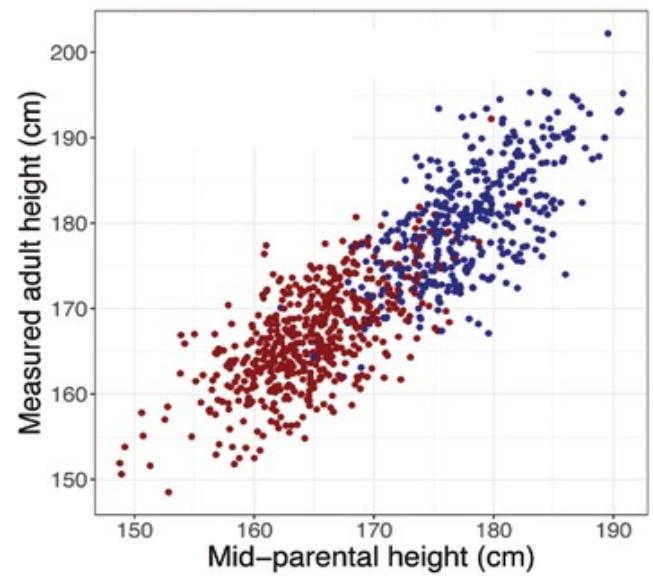
Necessary/Sufficient Conditions

- *Variation*
- *Heritability*
- *Differential Success*

A. Yes B. No C. …?







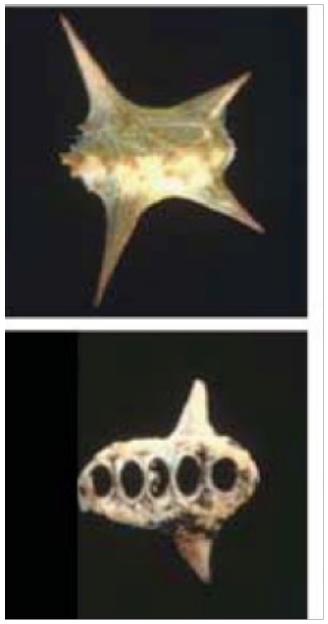
Thompson et al 2022

Discussion 4: Selection in Galápagos Finches

Medium Ground Finch, *Geospiza fortis*

Research Question:

- Does a major environmental disturbance (drought) drive selection in beak shape?



Absolute and Relative Fitness

Phenotype	Absolute Fitness	
	0	
	2	
	1	
	4	
	6	
	5	

Mean Absolute Fitness by Phenotype:

= (0+2+1)/3 =1

= (4+6+5)/3 = 5

Mean Absolute Fitness, Whole Population:

(0+2+1+4+6+5)/6 =3

Absolute and Relative Fitness

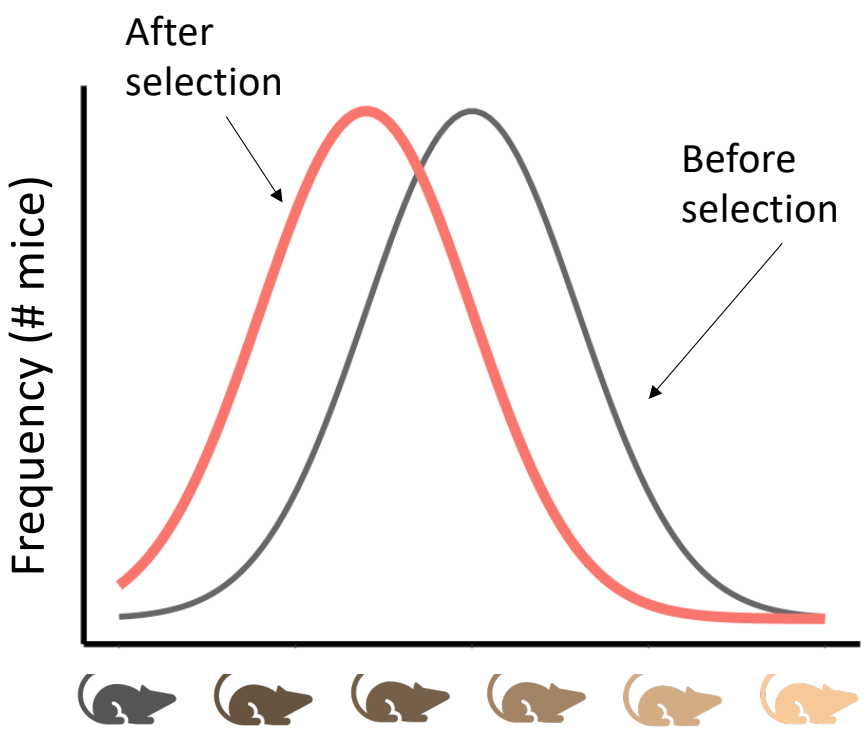
Phenotype	Absolute Fitness	Relative Fitness
	0	
	2	
	1	
	4	
	6	
	5	

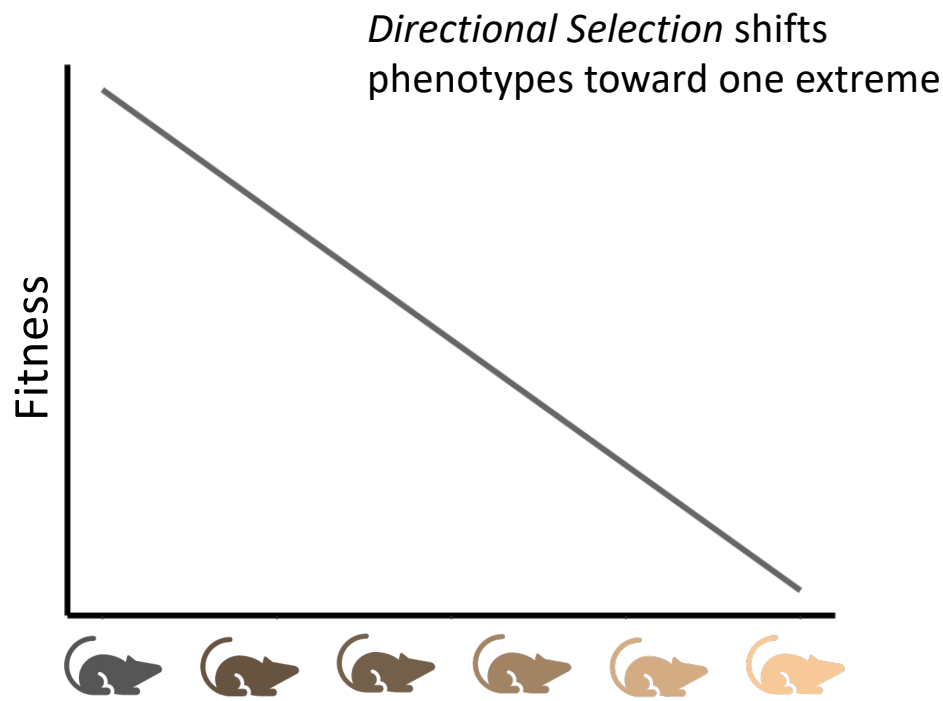
$$\textit{Relative fitness} = \frac{\text{Absolute Fitness}}{\text{Population Mean Fitness}}$$

Population mean fitness = 3

When is relative fitness > 1?
When is relative fitness < 1?

How does selection shift phenotypes?

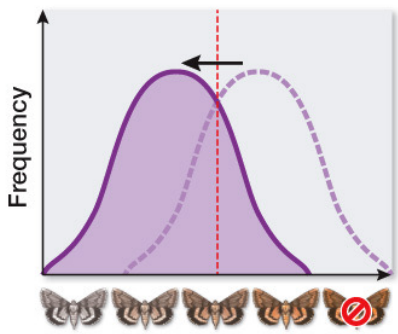




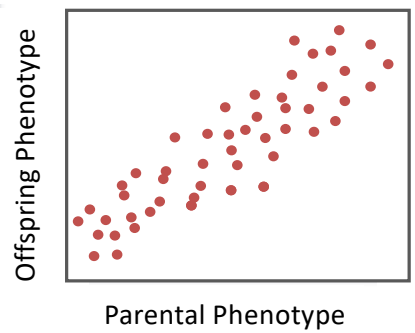
Predicting phenotypic change: The Breeder's equation

$$R = h^2 S$$

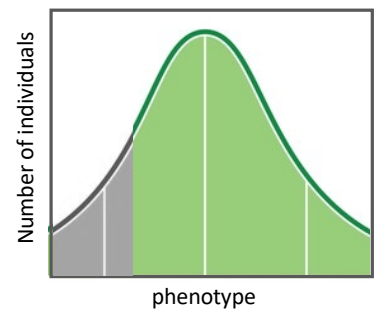
Response to selection: magnitude of phenotype shift between Generation 1 and Generation 2



Heritability: correlation between parental and offspring phenotypes

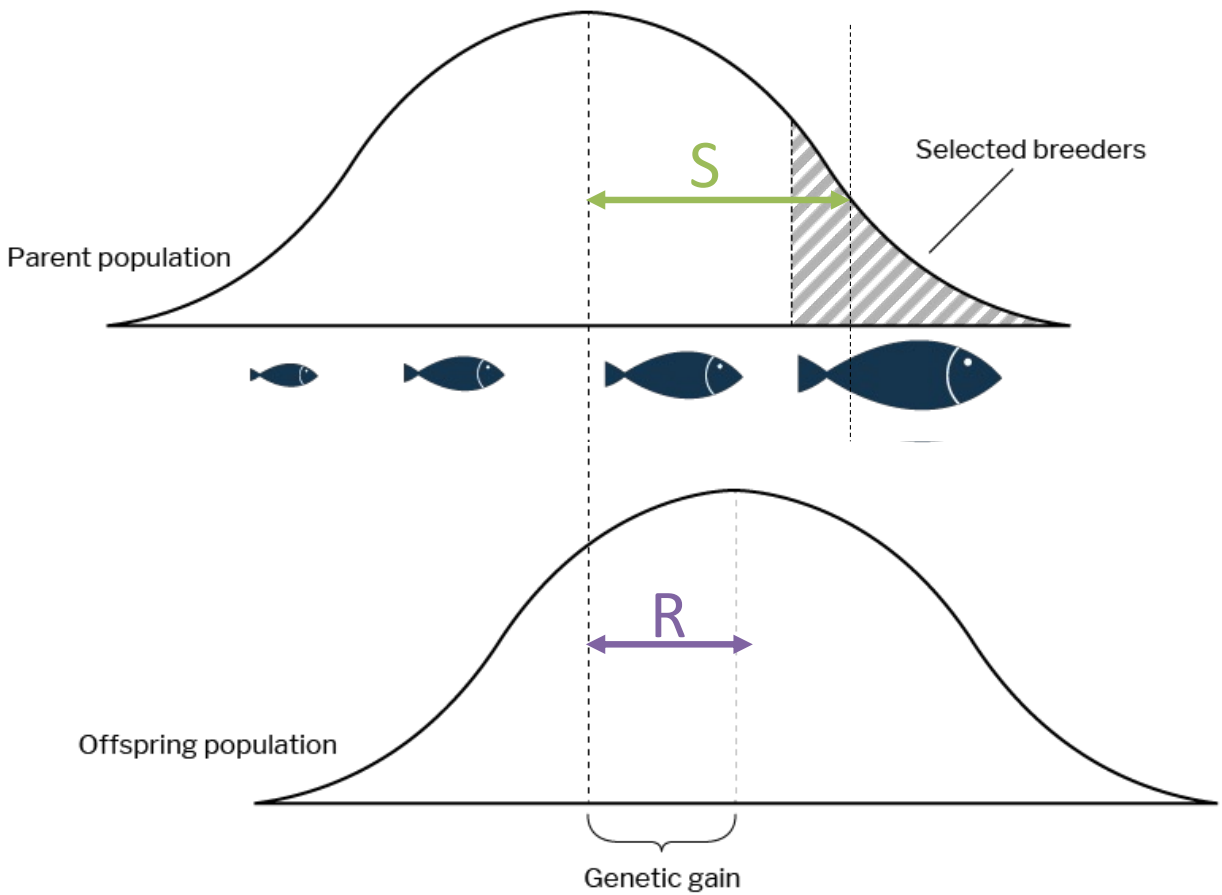


Strength of selection: magnitude of phenotypic difference between breeders and full population

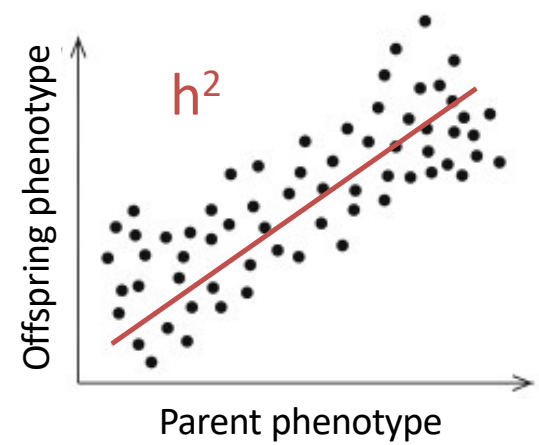


Note: heritability is h^2 not h
You do not need to square or square root any terms

Predicting phenotypic change: The Breeder's equation



$$R = h^2 S$$



Question

$R = h^2S$

The mean limb length for an island population of lizards if 1.1 cm. Lizards with longer limbs are better at clinging to branches and a hurricane sweeps away small-limbed lizards. After the hurricane, the mean limb length of surviving lizards is 1.6 cm.

What is the strength of selection (S)?

A. 1.1
B. 1.6
C. 0.5
D. 2.7
E. I'm not sure

Question

$R = h^2S$

The mean limb length prior to the hurricane was 1.1 cm. Lizards that survived the hurricane had mean limb length of 1.6 cm.

In the next generation, the mean limb length is 1.475 cm. What is h^2?

A. ~0.25
B. ~0.5
C. ~0.75
D. ~1
E. You can't calculate this without knowing R

Natural selection and altruism

Altruism: a behavior that reduces individual fitness and increases the fitness of other individuals

Vampire bats share blood meal with hungry bats

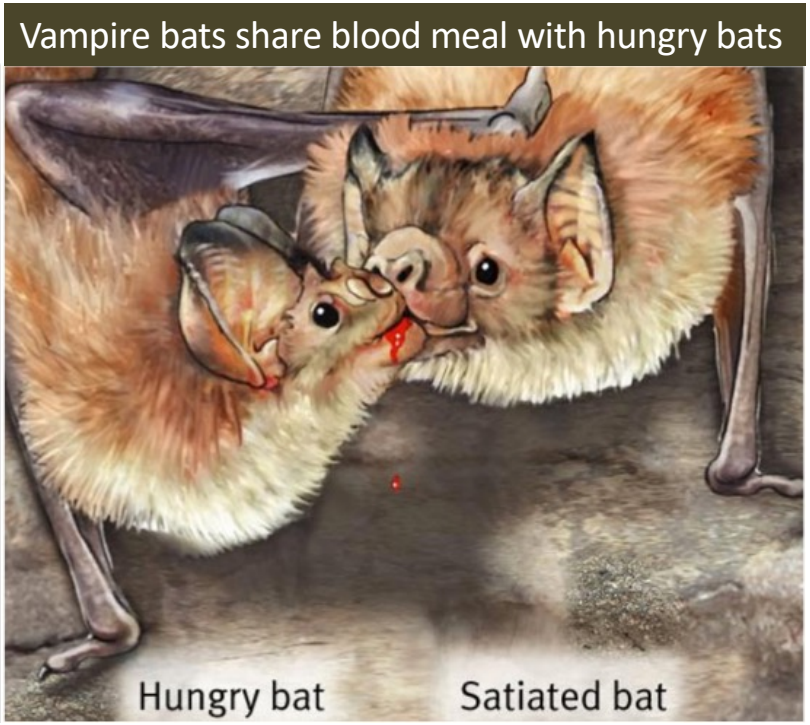
Only the honeybee queen reproduces

Belding ground squirrel alarm calls

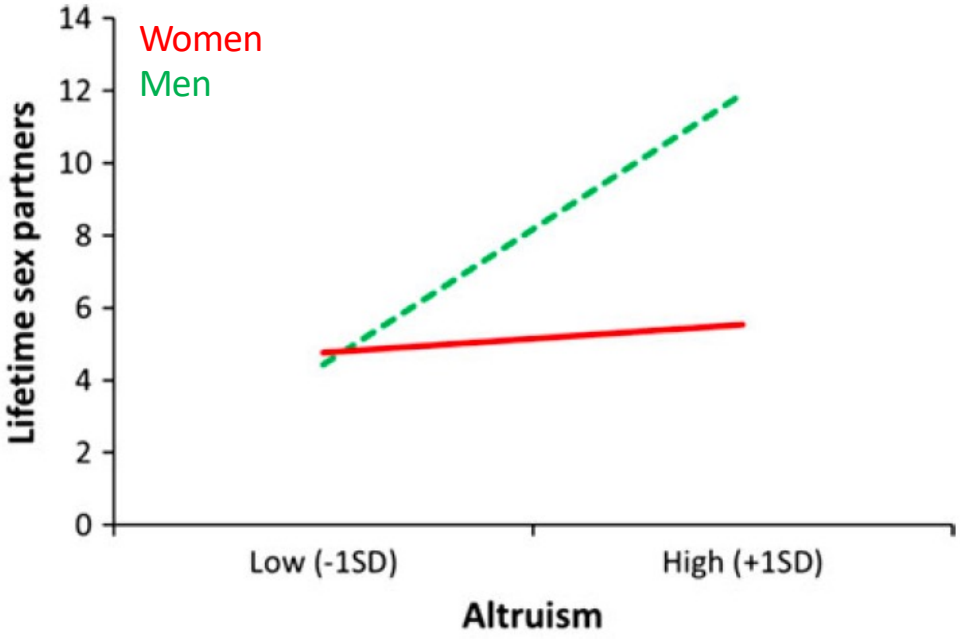
Naked mole rats help rear others' young

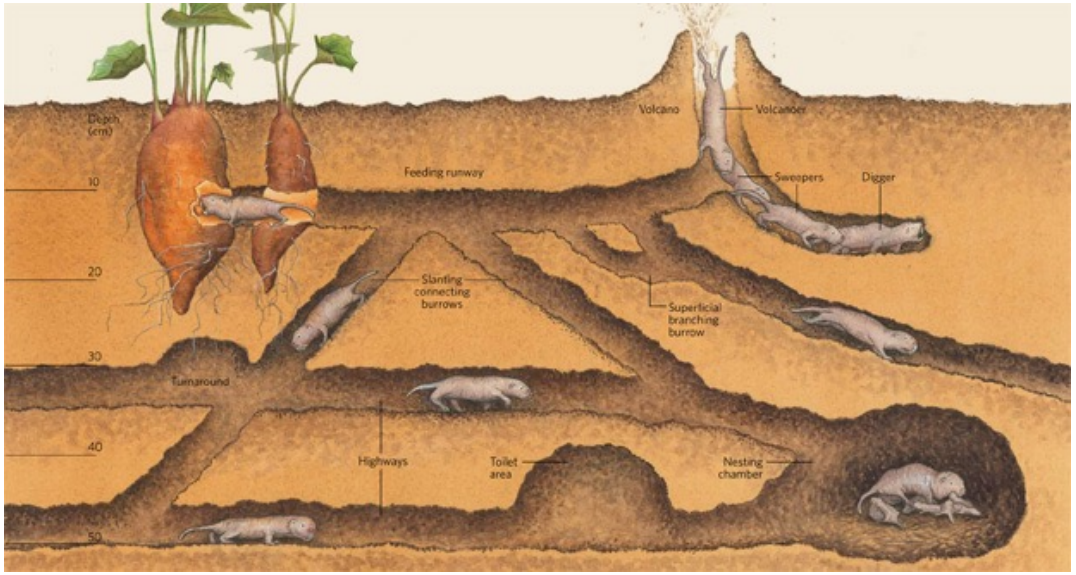
Direct benefits of altruism

Culture of reciprocation



Increased fitness



Indirect benefits of altruism

- Kin selection: selection that favors behaviors that increase the reproductive success of relatives
- Inclusive fitness: the sum of an individual's own fitness, and its contribution to the fitness of relatives